

FL Simulation Results

File: test_B1

Federated Learning Configuration

Total Clients (N): 10
Malicious Clients (M): 2
Neural Network: SimpleNN
Training Rounds: 15
Poisoned Rounds (R): 15
Distribution Strategy: last
Local Epochs (E): 3

Data Poisoning Attack Parameters

Poisoning Operation: Label Flip
Attack Intensity: 10.0% (0.10)
Poisoned Data Percentage: 100.0% (1.00)

Attack Summary: Using label_flip with 10.0% (0.10) intensity on 100.0% (1.00) of data for 15 rounds with 2 malicious clients.

Simulation Results

Init Accuracy: 0.7928
Clean Accuracy: 0.8723
Clean + DP Protection Accuracy: 0.8716
Poisoned Accuracy: 0.5649
Poisoned + DP Protection Accuracy: 0.8447
Drop (Clean - Poisoned): 0.3074
Drop (Clean - Init): 0.0795
Drop (Clean DP - Init): 0.0788
Drop (Poisoned - Init): -0.2279
Drop (Poisoned DP - Init): 0.0519
DP Protection Method: foolsgold
GPU Used: GPU 0

Confusion Matrix Metrics (Weighted Avg):

Clean Precision: 0.8783
Clean Recall: 0.8723
Clean F1 Score: 0.8717

Clean DP Precision: 0.8781
Clean DP Recall: 0.8716
Clean DP F1 Score: 0.8709

Poisoned Precision: 0.7594
Poisoned Recall: 0.5649
Poisoned F1 Score: 0.5734

DP Protection Precision: 0.8532
DP Protection Recall: 0.8447
DP Protection F1 Score: 0.8441

Summary

FL Simulation Complete
Task: a3782bb9-2a27-4cf0-b284-d49d29e9e029
GPU: GPU 0
Init Accuracy: 0.7928
Clean Accuracy: 0.8723
Clean DP Accuracy: 0.8716
Poisoned Accuracy: 0.5649
Data Poison Protection Accuracy: 0.8447
Drop (Clean - Poisoned): 0.3074
Drop (Clean - Init): 0.0795
Drop (Clean DP - Init): 0.0788
Drop (Poisoned - Init): -0.2279
Drop (Poisoned_DP - Init): 0.0519
Data Poison Protection Method: foolsgold
--- Confusion Matrix Metrics (Weighted Avg) ---
Clean Precision: 0.8783
Clean Recall: 0.8723
Clean F1 Score: 0.8717
Clean DP Precision: 0.8781
Clean DP Recall: 0.8716
Clean DP F1 Score: 0.8709
Poisoned Precision: 0.7594
Poisoned Recall: 0.5649
Poisoned F1 Score: 0.5734
DP Protection Precision: 0.8532
DP Protection Recall: 0.8447
DP Protection F1 Score: 0.8441